

JIANGYIFEI ZHU

Carnegie Mellon University ♦ CIC, 4720 Forbes Avenue, Pittsburgh, PA, USA, 15213

jiangyiz@andrew.cmu.edu

EDUCATIONS

Carnegie Mellon University

Pittsburgh, PA

Ph.D. in Electrical and Computer Engineering

May 2029

Advisors: Prof. Justin Chan & Prof. Swarun Kumar

Research Areas: Signal Processing, ML for sensor data, Wireless Sensing, Healthcare

Winner of the Qualcomm Innovation Fellowship (6% acceptance rate)

Carnegie Mellon University

Pittsburgh, PA

Master of Science in Electrical and Computer Engineering, GPA: 3.9/4.0

May 2024

Selected Coursework: Wearable Health Technologies, Computer Network, Real-time Embedded System

University of Massachusetts Amherst

Amherst, MA

Bachelor of Science in Computer Science, GPA: 3.8/4.0, **Dean's List**

February 2022

Bay State Scholar (A scholarship worth 50% of tuition)

Lewis & Clark College

Portland, OR

Coursework Completed Toward Bachelor's Degree

September 2017 - May 2019

PUBLICATIONS

- [Jiangyifei Zhu*](#), Yuzhe Wang*, Tao Qiang*, Vu Phan, Zhixiong Li, Evy Meinders, Eni Halilaj, Justin Chan, Swarun Kumar, **"GigaFlex: Contactless Monitoring of Muscle Vibrations During Exercise with a Chaos-Inspired Radar"**, *SenSys 26*
<https://arxiv.org/pdf/2511.13955>
- [Jiangyifei Zhu*](#), Kuang Yuan*, Akarsh Prabhakara, Yunzhi Li, Gongwei Wang, Kelly Michaelsen, Justin Chan, Swarun Kumar, **"Measuring multi-site pulse transit time with AI-enabled mmWave radars"**, *Nature Communications (under review)*
<https://arxiv.org/html/2510.18141v2>
- Siqi Zhang, Xiyuxing Zhang, Duc Vu, Tao Qiang, Clara Palacios, [Jiangyifei Zhu](#), Yuntao Wang, Mayank Goel, Justin Chan, **"LubDubDecoder: Bringing Micro-Mechanical Cardiac Monitoring to Hearables"**, *CHI 26*
<https://arxiv.org/abs/2509.10764>
- Yawen Liu, Akarsh Prabhakara, [Jiangyifei Zhu](#), Shenyi Qiao, Swarun Kumar, **"Shape-programming Robotic Reflectors for Wireless Networks"**, *IEEE ICRA 2025*

EXPERIENCES

Remote Muscle Monitor via Radar

Pittsburgh, PA

Advised by Justin Chan and Swarun Kumar

May 2025 - November 2025

Skills: Physiological Sensing, Radar Signal Processing, Chaos Theory, Deep Learning, Feature Engineering, User Study Design

- Engineered a full-stack FMCW mmWave radar pipeline, processing raw Intermediate Frequency (IF) signals to generate high-resolution Range-Doppler and Range-Angle maps for dynamic exercise monitoring.
- Pioneered a physics-aware signal processing framework utilizing Chaos theory on raw phase signals to model aperiodic micro-Doppler vibration dynamics, effectively isolating subtle (1–100 Hz) muscle signatures from gross body motion.
- Designed a dual-branch deep learning architecture with Multi-Scale Temporal Feature extraction and SE-Attention to fuse subject-specific embeddings, achieving robust vibration modeling with ~6% error.
- Developed an interpretable fatigue prediction module using Logistic Regression on extracted phase-space features; achieved 0.86 AUC in classifying high-fatigue states (1-RIR) vs. standard repetitions.

Contactless Multi-Site Pulse Sensing by Radar

Pittsburgh, PA

Advised by Justin Chan and Swarun Kumar

September 2023 - March 2025

Skills: Physiological Sensing, Radar Signal Processing, Adaptive Filtering, Deep Learning, User Study Design

- Implemented a high-resolution mmWave signal processing pipeline using Range FFT and beamforming on an 86-virtual-antenna array (1.4° angular resolution) to isolate minute physiological signals ($\approx 30\mu\text{m}$) from major noise sources like respiratory motion (1-4 mm)
- Implemented an adaptive periodicity-driven algorithm that ranks spatial bins via autocorrelation, enabling robust signal selection across the heart, neck, and wrist despite multi-path interference and clothing occlusion.
- Contributed to the design and implementation of a multi-branch Deep Neural Network architecture, featuring bi-directional LSTMs for temporal modeling and a Cross-region Attention Fusion Module, to jointly and accurately estimate multi-site Pulse Transit Time (PTT) and Diastolic Blood Pressure (DBP). The system achieved clinical-grade performance (DBP R=0.92, PTT R=0.89) and met FDA AAMI guidelines.

Software Engineer @ Huawei Digital Technologies Co., Ltd

Beijing, China

Cluster Computer Solution Team

March 2022 - August 2022

Skills: PyTorch, Hardware Acceleration, System Optimization

- Engineered custom PyTorch operators for Graph Neural Networks (GNNs) optimized for heterogeneous hardware architectures (Huawei Ascend NPUs, Kunpeng ARM, Intel x86), enabling high-performance cross-platform model inference.
- Designed a dynamic computing resource allocation strategy (pre-partitioning) for the cluster scheduler, slashing environment deployment latency by over 95% (from 30 min to <1 min).
- Developed an automated benchmarking pipeline to profile model performance and operator latency across different hardware backends, eliminating significant manual overhead (~4 hours saved per test cycle).

SKILLS

Programming: Python, MATLAB, C/C++, Java, Dart/Flutter, JavaScript, C#

ML & Signal Processing: Deep Learning (PyTorch/TensorFlow), Time-series & Chaos features, Radar/IMU/sEMG/PPG processing, Adaptive filtering & Periodicity analysis,

Software & Tools: Linux, Git, Docker, LaTeX, mobile & Web development

Sensing/Hardware: TI mmWave Radar (AWR series), PPG, IMU, Embedded Systems, Arduino, DAQ Systems.

SCHOLARSHIPS

- Qualcomm Innovation Fellowship Winner May 2025
Selected for the proposal "*Contactless Cardiovascular Health Monitoring using AI-enabled mmWave Radars*"
- Center for Machine Learning and Health Fellowship at Carnegie Mellon University April 2025

SERVICES

Reviewer: CHI 2026, IEEE ICC 2026 (IoT Symposium), ACM IMWUT 2025

TALKS

Guest Lecturer, "Advanced Topics on Wireless Technology" at CMU Nov 2024, Sep 2025, Nov 2025

Course: 18-441/18-741 Computer Networks; 17320/17720, Machine learning and sensing for healthcare

Guest Presenter, "Detect Like a Superhero: Using Waves to Sense!", at CMU Apr 2025

K-12 Outreach via CMU Gelfand Center